

D03. Solution Architecture and Component Design

Structure Overview

The solution is organised into a small number of core components, each with a clear technical role in the simulation of the selected inbound warehouse process. These components consist of Linux/Bash scripts, Oracle SQL, a small C++ element, configuration files and validation outputs. Together, they support the setup, execution, data handling, and checking of the selected flow in a clear and modular way.

- **Linux/Bash Scripts** are used to manage the execution of the simulation, including setup, runtime control, logging, and validation.
- **Oracle SQL** is responsible for managing, reading and manipulating data produced in the warehouse environment.
- **C++** contributes to the logic by validating data, applying rules, interacting with the system as defined by the configuration files and the operational flow.
- **Configuration files** define the structure, docks, assignment rules, and operational settings of the simulated warehouse before execution begins.
- **Validation outputs** provide technical evidence used to confirm that the simulation has produced the expected result.

How the Components Interact

The simulation is organised to support the selected inbound flow. It starts with the configuration files, which define the structure, rules, and main operational settings of the simulated warehouse environment. The Linux/Bash scripts then use these settings to prepare the runtime environment and run the inbound process step by step. During execution, the scripts work with the Oracle SQL components to store, query, and modify warehouse objects and sample data, while PL/SQL is used to implement selected automated logic. A small C++ element is also included to perform one focused task, such as applying a simple warehouse rule, supporting location ranking, or carrying out rule-based validation. Once the flow is completed, the system generates validation outputs, including logs and snapshots, which are used to confirm that the simulation produced the expected final result and that the inbound process was executed correctly.